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A Theoretical Study on Redox Potential and pK_a of [2Fe-2S] Cluster Model from Iron-Sulfur Proteins

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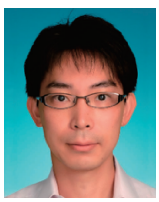
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Abstract

Iron-sulfur proteins are essential in several biological processes, such as electron transfers in photosystems and respiratory chains. The range of redox potentials of these proteins are so wide that many substrates can interact with them. Thus, the redox potentials of the iron-sulfur proteins are crucial to facilitate the process. This study aims to investigate the redox potential of iron-sulfur proteins with [2Fe-2S] core by using two typical exchange-correlation functionals in density functional theory (DFT), i.e., B3LYP and M06, with and without the diffuse function. Since it involves transition metals, several types of spin approximation were also explored, such as high spin (HS), low spin (LS), approximated spin projection (AP), and J coupling parameter (JC) methods. We found that the diffuse function contributes to improving the accuracy. We also found that M06 functional produced more accurate results than B3LYP. The best result is obtained for AP UM06/6-31++G(d,p) with the absolute error of 0.01 V.

Keywords: Redox potential | [Fe2S2] cluster model | Density functional theory (DFT)

1. Introduction

Iron-sulfur proteins are commonly found in living organism and are involved in different processes in biological systems, such as nitrogen fixation, photosynthesis, and respiration.[1–3] These proteins have a wide range of reduction potentials that enable them to interact with diverse types of redox substrates by acting as electron carriers.[4,5] They contain one or more clusters that are composed of one to eight iron atoms, cysteine sulfur atoms, and inorganic acid labile sulfurs that act as bridging atoms. To promote biological processes, these clusters play a vital role in various functions such as radical chain stabilization,[1] redox reaction,[4,5] and Lewis acids.[6] The iron-sulfur clusters are classified by the number of iron and sulfur atoms, structural motif, and its electrochemical and spectroscopic properties.[2] Although the structures of iron-sulfur

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clusters in different iron-sulfur protein are quite similar, the residues interact with the cluster are different. Thus, the environmental effect around the iron-sulfur cluster is very important in knowing its function.

Among several classes, ferredoxins are iron-sulfur proteins that have low molar mass and function as electron carriers in various biological pathways.[7] These proteins were confirmed by EPR signals from a nonheme iron.[8,9] As a iron-sulfur protein, ferredoxins have a wide range of redox potentials that lead the ability to interact with a diverse type of redox substrate in several vital processes in biological systems.[2,4,5,10]

Ferredoxins are further classified into several classes. One of the classes is [2Fe-2S] ferredoxin containing a complex cluster of two iron atoms, two inorganic sulfur atoms, and four cysteine thiolates. Since ferredoxins have attracted much interest for their role in many biological processes, vast numbers of experimental studies have been devoted to exploring structure-function relationships of these proteins.[10–13] A crystal structure of a clostridium ferredoxin was firstly obtained by Tagawa and Arnon in 1962, followed by a measurement of the redox potential.[12] The presence of inorganic sulfur in the iron-sulfur cluster was confirmed by Buchanan et al. where the amount of sulfur was found to be equimolar with the non-heme iron.[11] The redox potentials and stoichiometry of electron transfer in ferredoxins were reinvestigated by Tagawa and Arnon.[13] Their studies revealed that the redox potential of spinach ferredoxin is about -420 mV.

Following experimental studies, several theoretical studies of ferredoxins were performed to complement understanding of this protein.[14–18] Noodleman et al. investigated various clusters identical to the active site of ferredoxin by performing X valence bond scattered wave (X-VB-SW) calculations.[14] They found that the bonds between iron atoms and inorganic bridging sulfur were stronger than the bonds between the iron and sulfur of thiolate. Noodleman et al. continued their study by utilizing broken symmetry analysis to estimate antiferromagnetic coupling constants in iron-sulfur clusters.[15] Although their calculations significantly overestimated the corresponding experimental data, the trend of the results is in agreement with them indicating the validity of the method.

Mouesca et al. calculated the redox potential of the iron-sulfur cluster by considering several parameters such as energies of ground state, spin Hamiltonian parameter, and resonance delocalization parameter. Those parameters were approximated from broken symmetry and high spin state calculations.[16] They produced a result in good agreement with experimental data. They also introduced the concept of spin and electron delocalization barycenter analogous to the concept of ligand field stabilization theory. Banci et al. analyzed the factors that determine the magnitude of the redox potential of the ferredoxin. Their results confirmed that the net charges of acidic and basic groups are responsible for the wide range of the reduction potentials.[17]

Recently, we have proposed an accurate estimation method for the redox potentials of organic and inorganic materials by means of electronic structure calculations with solvation models.[19–21] Since the solvation models often fail to estimate solvation free energies of highly charged ions, conventional methods over or underestimate the redox potentials of transition metal complexes with $+3/-3$ charges. Later, we succeeded in reproducing the redox potential of transition metal complexes within 0.25 V by adding correction terms originating from counter ion around the metal complexes referred as a Pseudo Counter Ion Solvation Scheme (PCIS).[20] The protein environment is often modeled by a dielectric medium with $\varepsilon = 4$–10 and a discrepancy between results by the solvation model and those by real models show the importance of the actual protein environment. Thus, it is quite interesting to estimate redox potentials of metalloproteins such as heme proteins, iron-sulfur proteins, and copper proteins by using the present method and assess the performance of it.

In this study, we aim to calculate the redox potential of two [2Fe-2S] ferredoxins, i.e., spinach ferredoxin and adrenodoxin. We here analyze relative values of the redox potentials between two proteins instead of absolute ones to reduce the systematic error. In particular, we investigate the dependence of the redox potential of these ferredoxins on two DFT functionals, B3LYP and M06, and two basis sets w/wo diffuse functions. Especially M06, which was originally proposed by Zhao and Truhlar,[22,23] is known to give better results than B3LYP in the calculation of organic and inorganic molecules.[24–26] The models that are presented in this study contain an [Fe2S2] iron-sulfur cluster, which has several spin states. Thus, we utilized four approximations related to this issue as mentioned below. The solvation contribution was accounted by using an implicit solvent model. We summarize the combination of these methodologies and approximation methods for the performance of the redox potential of [Fe2S2]. In addition, we also investigated the singly occupied natural orbital (SONO) and atomic partial charge of the cluster in reduced state to understand the difference in the redox potentials between two ferredoxins.

2. Computational Methods

The calculations were conducted by using two types of hybrid exchange-correlation functionals, i.e. B3LYP and M06, with two basis sets, i.e. 6-31G(d) and 6-31G(d,p), w/wo diffuse functions. All calculations were performed using the Gaussian 09 program package.[27] As mentioned in the introduction, two iron-sulfur proteins, i.e. ferredoxin and adrenodoxin, were investigated in this work. The structures of cluster models were taken from X-ray crystal structures with PDB ID 1A70 (ferredoxin), 2MJD and 2MJE (adrenodoxin). Four cysteine residues in the cluster were replaced by methylthiolates. The hydrogen atoms were optimized by using B3LYP/6-31G(d) level of theory (see Figure 1) while the other heavy elements were fixed.

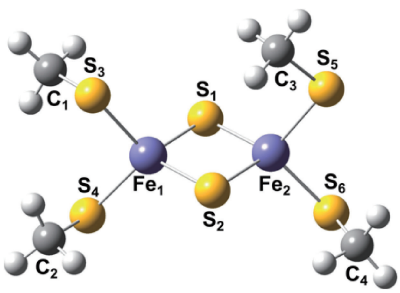


Figure 1. The structure of iron-sulfur cluster model.

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Table 1. The formula of coupling constants and ground state energy.[18]

Oxidized State	Reduced State
$S_1 = S(Fe^{3+}) = 5/2$	$S_1 = S(Fe^{3+}) = 5/2$
$S_2 = S(Fe^{3+}) = 5/2$	$S_2 = S(Fe^{2+}) = 2$
	$B = 1/10[e(HS)_u - e(HS)_g]$
$J_{ox} = 2/25[E_0(HS) - E_0(BS)]$	$J_{red} = 1/10[E_0(HS)_g - E_0(BS) + 5B]$

$e(HS)_u$ and $e(HS)_g$ are orbital energies belong to u and g component of metal d orbitals split by resonance delocalization.

The spin states of the clusters were approximated by several spin approximations. We consider the spin state of the clusters as high spin (HS) and low spin (LS), in which the energy of LS state was approximated by using broken symmetry (BS) technique.[28] As for the LS states, we also approximate the energy by using approximate spin projection (AP) method as[29–31]

$$E^{LS}_{APBS} = \alpha E^{LS}_{BS} - \beta E^{HS}, \quad (1)$$

where

$$\alpha = \frac{\langle \hat{S}^2 \rangle^{HS} - \langle \hat{S}^2 \rangle^{LS}_{exact}}{\langle \hat{S}^2 \rangle^{HS} - \langle \hat{S}^2 \rangle^{LS}_{BS}}, \quad (2)$$

$$\beta = \alpha - 1. \quad (3)$$

Meanwhile, another approximation was also used for the energy of LS state. In this approximation, we consider J spin coupling (JC) in a Heisenberg Hamiltonian with an isotropic correction, with the ground state energy expressed as[16,18]

$$E_0(GS) = E_0(BS) - 5/2J_{ox}, \quad (4)$$

$$E_0(GS) = E_0(BS) - 2J_{red}, \quad (5)$$

where J represents the spin coupling parameter. We used the parameter of the Heisenberg Hamiltonian obtained from the reference,[16,18] as shown in Table 1. The parameter of S_1 and S_2 represent spin eigenvalues on two iron centers, B represents resonance delocalization parameter, and S represents total spin.

Conductor-like polarizable continuum model (CPCM), with $\varepsilon = 10$, was used to account for solvent contribution around the cluster.[32,33] We also added a correction term in calculating the solvation energy based on PCIS.[20] This scheme includes charge-dependent correction terms, which are derived from a Generalized Born theory. This correction scheme can reduce an error in the redox potential for highly charged metal complexes.[20] Standard redox potential (E_{redox}) was calculated by using the relationship with the Gibbs free energy difference between oxidized and reduced states (ΔG_{redox}) as

$$E_{redox} = \frac{\Delta G_{redox}}{nF} - E_{SHE}, \quad (6)$$

where n is number of electrons, F is Faraday constant, and E_{SHE} is absolute potential of the standard hydrogen electrode, taken as 4.43 V.[34] In this study, the free energy in the protein environment was directly calculated by CPCM instead of by a thermodynamic cycle. The free energy difference was approximated as the energy difference between the oxidized and reduced states, i.e., the vertical ionization potential (VIP), as

$$\Delta G_{redox} \sim E^{ox}_{(aq)} - E^{red}_{(aq)}. \quad (7)$$

The free energy correction terms were completely neglected owing to high computational costs required for vibrational analyses. However, we experienced previously that this approximation works well in estimating the redox potentials of metal complexes.[20]

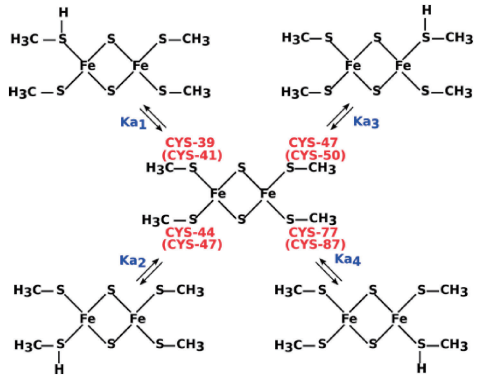


Figure 2. The model of protonation of cysteine residue in iron-sulfur cluster in ferredoxin (adrenodoxin).

The acid dissociation constant (pK_a) was calculated by using free energy difference upon a deprotonation reaction (ΔG_{dep}) as

$$pK_a = \frac{\Delta G_{dep}}{2.303\,RT}, \quad (8)$$

where R and T are gas constant and temperature, respectively. The calculation of free energy was derived from deprotonation reaction (see scheme in Figure 2), $AH \Leftrightarrow A^- + H^+$, as

$$\Delta G_{dep} = E_{aq}(A^-) + G_{aq}(H^+) - E_{aq}(AH), \quad (9)$$

where the free energy correction terms were again neglected in a similar way with redox potential calculation. Then, the free energy of a proton in aqueous phase was calculated as

$$G_{aq}(H^+) = G_g(H^+) + \Delta G_{aq,solv}(H^+) + \Delta G^{1\,atm \to 1\,M} \quad (10)$$

where $\Delta G_{aq,solv}(H^+)$ is taken as -265.9 kcal/mol.[35] The $G_g(H^+)$ is equal to -6.28 kcal/mol at 298.15 K, as calculated using Sackur-Tetrode equation for gas-phase monoatomic species.[36] The $\Delta G^{1\,atm \to 1\,M}$ term was added to convert a standard state of 1 atm to 1 M, which the value is taken as 1.89 kcal/mol.[37] According to the protonation models in Figure 2, we calculated the energy difference of iron-sulfur cluster in the oxidized state and protonated reduced state. This value will represent the feasibility of the iron in each site to be protonated.

Meanwhile, we also perform molecular orbital (MO) analysis to investigate the orbital shape of the SONO orbital of the iron-sulfur cluster in a reduced state. This analysis was performed by using localized natural orbital method (LNO).[38] The atomic partial charges of the clusters were also calculated by using Merz-Singh-Kollman scheme.[39]

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3. Results and Discussion

At first, we discuss the reliability of each method to reproduce the experimental results. From the experimental study, the redox potential of spinach ferredoxin and the adrenodoxin were known as −0.42 V[13] and −0.27 V,[40] respectively. In this study, we discuss the absolute error derived from relative redox potential between those proteins, i.e $\Delta E = E_{\text{ferredoxin}} - E_{\text{adrenodoxin}}$, instead of absolute redox potentials to dismiss systematic error. The absolute errors for relative redox potential toward the experimental results were defined as the difference between relative error obtained from calculated and experimental study ($\Delta\Delta E = |\Delta E_{\text{calc}} - \Delta E_{\text{exp}}|$). Judging from these values, we estimate the reliability of combinations of methodologies. The absolute value of redox potential of the iron-sulfur cluster is presented in Table S1 in supporting information.

The results of $\Delta\Delta E$ are summarized in Table 2. In the case of B3LYP calculation without PCIS scheme, we found that diffuse function did not give any improvement on $\Delta\Delta E$ in any spin approximations, except for JC approximation. $\Delta\Delta E$ values by HS, BS, and AP UB3LYP w/wo diffuse function were similar, being 0.23, 0.21, and 0.20 V, respectively. On the other hand, those for the JC approximation w/wo diffuse function were 0.26 and 0.21, respectively. This indicates that JC approximation may improve them a bit by adding the diffuse function, though the improvement does not become remarkably close to the corresponding experimental result. This improvement is caused by the method of JC approximation that defines the Hamiltonians for the oxidized and reduced states in a different way. In contrast, the diffuse function gives a significant improvement to $\Delta\Delta E$ when one adopts M06 functional with AP approximations, i.e., from 0.35 to 0.03 V.

In VIP with PCIS scheme, the basis set dependence of B3LYP functional is again little. However, the inclusion of the diffuse function gives an improvement in $\Delta\Delta E$ with BS and JC UM06 levels of theory. The most dramatic improvement due to the diffuse function was found in the calculation with AP UM06, where the absolute error decreases by 0.30 V (from 0.31 to 0.01 V). These findings suggest the importance of diffuse function in improving the calculation of redox potential of the iron-sulfur cluster when one adopts the M06 functional.

Table 2. The absolute error of redox potential difference.

Method	Absolute Error			
	HS	BS	AP	JC
VIP Scheme				
1	0.23	0.21	0.20	0.26
2	0.23	0.21	0.20	0.21
3	0.23	0.28	0.35	0.18
4	0.22	0.13	0.03	0.12
VIP with PCIS scheme				
1	0.19	0.17	0.16	0.22
2	0.19	0.17	0.16	0.17
3	0.19	0.24	0.31	0.22
4	0.18	0.09	0.01	0.08

1: B3LYP/6-31G(d), **2**: B3LYP/6-31++G(d,p), **3**: M06/6-31G(d), and **4**: M06/6-31++G(d,p).

Regarding the DFT exchange-correlation functional, we found that the M06 functional gives more accurate results than B3LYP functional, except for HS and AP approximations without diffuse function. In VIP with PCIS scheme with BS approximation, $\Delta\Delta E$ by B3LYP and M06 with diffuse function are 0.17 and 0.09 V, respectively. This difference is related to two hybrid characteristics of those functionals, the ratio of exact HF exchange functional and the type of exchange functional. For the latter issue, meta-GGA functional is used to couple with exact HF exchange in M06, while pure GGA functional is used in B3LYP functional. Overall, the result with the best accuracy was obtained for AP UM06/6-31++G(d,p) with the absolute error of 0.01 V.

Since the calculations were performed using an implicit solvent model, we here figure out a correlation of the absolute error to the inverse of ε, i.e., $1/\varepsilon$. According to Figure 3, we found that the absolute value of the redox potential was inversely proportional to the dielectric constant, for which the tendencies of B3LYP and M06 were almost similar. However, the experimental results can be reproduced for neither the spinach ferredoxin (−0.42 V) nor adrenodoxin (−0.27 V) within a range from $\varepsilon = 1$ (vacuum) to $\varepsilon = 78$ (water). This means that the present model does not mimic the protein environment well.

To obtain a more reliable result, we should include the huge protein environment in the QM cluster model. Although we have tested a QM/MM model with small [Fe2S2] cluster treated here, the obtained results were worse than the present simple cluster model, where the absolute error is more than 2 (see Table 2 in supporting information). In future work, we consider the cluster size effects on the absolute value of the redox potential. We also depicted the dependency of $\Delta\Delta E$ on the dielectric constant as shown in Figure 3. Although $\Delta\Delta E$ increases a bit with increase of $1/\varepsilon$ value, the redox potential difference between the ferredoxin and adrenodoxin is almost a constant value unlike the absolute values, indicating less importance of the dielectric constant for $\Delta\Delta E$.

We next confirmed the possibility of other protonation states of the [Fe2S2] cluster by estimating pK_a values of thiol groups in cysteine as shown in Table 3. Since pK_a values were found to be larger than 7, these residues remained deprotonated upon reduction in vivo. There is little methodology dependence for the ferredoxin. On the other hand, a large methodology

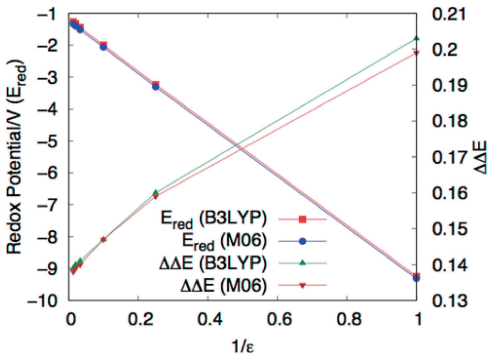


Figure 3. The dependency of absolute redox potential of ferredoxin in cluster model and $\Delta\Delta E$ as a function of inverse dielectric constant by B3LYP and UM06/6-31++G(d,p).

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Table 3. The calculated pK_a value for ferredoxin (adrenodoxin).

Method	pK_{a1}	pK_{a2}	pK_{a3}	pK_{a4}
1	15.7 (19.1)	15.0 (20.5)	16.3 (21.4)	14.7 (20.3)
2	15.0 (18.1)	15.4 (20.1)	16.0 (19.0)	14.2 (18.3)
3	14.6 (17.1)	14.0 (17.2)	14.6 (17.3)	12.6 (17.1)
4	14.2 (17.8)	14.1 (18.7)	15.3 (17.3)	12.9 (17.1)

1: B3LYP/6-31G(d), **2**: B3LYP/6-31++G(d,p), **3**: M06/6-31G(d), and **4**: M06/6-31++G(d,p).

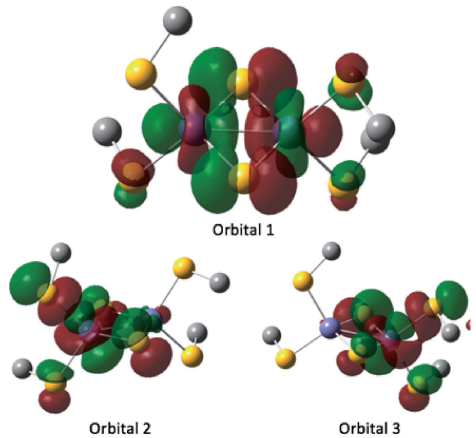


Figure 4. Three of nine singly occupied natural orbitals (SONO) of iron-sulfur cluster in reduced state. The hydrogen atom was neglected for clarity reason.

dependence was found for the adrenodoxin. The diffuse function contributed to the pK_a values by B3LYP and M06 calculations in a different manner. Generally, the inclusion of the diffuse function decreases (increases) the pK_a values by B3LYP except for pK_{a2} (M06 except for pK_{a1}). We also found that pK_{a4} of the ferredoxin exhibits the lowest pK_a among four different S sites in all methods and pK_{a2} of the adrenodoxin does for methods **2** and **4**. This is due to almost identical configurations, but slightly deformed structures of the [Fe2S2] clusters in both proteins, which also affect the absolute redox potentials of these [Fe2S2] proteins.

To confirm the site dependency of pK_a and the difference in the redox potentials between two proteins, a singly occupied natural orbital (SONO) of the iron-sulfur cluster in a reduced state is depicted in Figure 4. Since we are concerned with the iron-sulfur cluster in the ferromagnetic state, we have identified nine SONO orbitals. Those orbitals consist of one orbital representing a mix of d_{z^2} orbital between both iron atoms and each four orbitals localized around Fe^{2+} and Fe^{3+} atoms. We present three of nine SONO orbitals depicted in Figure 4. The remaining six SONO orbitals are available in supporting information (Figure S3). In orbital 1, we found four bonding interactions in d_{z^2} orbitals of irons and σ orbitals of sulfide ions. Besides, each iron ion also interacts with two σ orbitals of sulfur atoms in cysteine residues by anti-bonding interaction. This finding confirms that the covalent bonding between iron and sulfide ion is stronger than the bonding between iron and the sulfur atom of cysteine. We also found the d_{z^2} orbitals that localized in each iron atom as represented in orbital 2 and 3.

Table 4. The atomic partial charge of ferredoxin (adrenodoxin) in reduced state. The atom number refer to Figure 1.

Atom	Method			
	1	**2**	**3**	**4**
Fe_1	0.98 (1.19)	1.06 (1.64)	0.94 (1.16)	0.97 (1.55)
Fe_2	0.81 (1.11)	0.87 (1.43)	0.76 (1.06)	0.79 (1.32)
S_1	−0.92 (−1.03)	−0.91 (−1.14)	−0.90 (−1.01)	−0.86 (−1.09)
S_2	−0.92 (−1.06)	−0.91 (−1.14)	−0.90 (−1.04)	−0.86 (−1.09)
S_3	−0.81 (−0.89)	−0.90 (−1.12)	−0.79 (−0.87)	−0.85 (−1.07)
S_4	−0.78 (−0.86)	−0.87 (−1.05)	−0.76 (−0.84)	−0.83 (−1.03)
S_5	−0.76 (−0.85)	−0.78 (−1.00)	−0.74 (−0.83)	−0.74 (−0.96)
S_6	−0.74 (−0.85)	−0.81 (−1.05)	−0.71 (−0.83)	−0.77 (−1.03)
C_1	0.05 (0.08)	0.08 (0.15)	0.03 (0.07)	0.06 (0.13)
C_2	0.04 (0.08)	0.08 (0.17)	0.03 (0.03)	0.06 (0.16)
C_3	0.03 (0.00)	0.01 (−0.03)	0.02 (−0.01)	0.00 (−0.04)
C_4	0.03 (0.09)	0.05 (0.14)	0.01 (0.08)	0.03 (0.14)

1: B3LYP/6-31G(d), **2**: B3LYP/6-31++G(d,p), **3**: M06/6-31G(d), and **4**: M06/6-31++G(d,p).

The atomic partial charges of the iron-sulfur cluster in the ferredoxin and the adrenodoxin are listed in Table 4. As expected, the charge of Fe_1 (Fe^{3+}) atom is higher than the charge of Fe_2 (Fe^{2+}) atom. We also found identical charges of both sulfide ions, which confirm an almost homogenous charge distribution on both ions. The averaged charge on S_3 and S_4 atoms is found to be larger than that on S_5 and S_6 atoms because they bind to difference charges of iron atoms. This leads to heterogeneity of charge distribution on them. Regarding the calculation method, we found a remarkable dependence of the exchange-correlation functional and diffuse function on the atomic partial charges, especially on iron atoms. The existence of d orbitals in iron leads to the sensitivity of the calculation method to atomic partial charges. The implementation of diffuse function increases the partial charge of atoms with positive charges, and decreases the partial charge of atoms with negative charges. This indicates that more polarization of the electron density was induced owing to the diffuse function.

Regarding the results of the absolute error of redox potential shown in Table 2, we found that the absolute error produced by AP M06/6-31G(d) is larger than the other AP calculations. We can relate these results with the tendency of partial atomic charges shown in Table 4. The atomic partial charges of non-

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carbon atom calculated by M06/6-31G(d) are found to be underestimated compared to the other methods. Actually, it is also reported that the error of the M06 method for reduction potential of FeC is worse compared to the M11 and other exchange correlation functionals owing to the description of Fe atom.[41] Thus, the choice of the exchange correlation functional and basis set dependence are of great importance for the investigation of FeS clusters.

4. Conclusion

We have investigated the redox potential and pK_a of cluster models in iron-sulfur protein. We found the contribution of diffuse function in improving the results, which indicated by the decreasing of absolute error ($\Delta\Delta E$). Meanwhile, M06 produces more accurate results compared to the B3LYP calculation. Regarding the solvation scheme, an improvement of the results was obtained from the implementation of PCIS scheme in VIP approximation. Amongst all methods, the best result was obtained from the calculation by AP UM06/6-31++G(d,p) with the absolute error of 0.01 V.

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Supporting Information

The supporting information contains the absolute redox potential of both cluster and hybrid models, other six SONO orbitals, and the cartesian coordinate of all cluster models. This material is available on http://dx.doi.org/10.1246/bcsj.20180116.

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